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Required Technology Stack - Learning Curve			
Order of Precedence	Name	Description	Links
1	GitHUB	**GitHub**, Inc. is a United States-based global company that provides hosting for software development version control using Git.	Team Confluence Support Page: GitHub https://github.allstate.com/
2	Gradle	**Gradle** is an open-source build automation system that builds upon the concepts of Apache Ant and Apache Maven and introduces a Groovy-based domain-specific language instead of the XML form used by Maven for declaring the project configuration.	Team Confluence Support Page: GRADLE
3	Jenkins Pipeline	**Jenkins Pipeline** (or simply "**Pipeline**") is a suite of plugins which supports implementing and integrating continuous delivery **pipelines** into **Jenkins**. ... The definition of a **Jenkins Pipeline** is typically written into a text file (called a Jenkinsfile) which in turn is checked into a project's source control repository.	Team Confluence Support Page: Jenkins Pipeline https://login.cf.nonprod-mpn.ro11.allstate.com/login
4	SonarQube	**SonarQube** is an open-source platform developed by **SonarSource** for continuous inspection of code quality to perform automatic reviews with static analysis of code to detect bugs, code smells, and security vulnerabilities on 20+ programming languages.	Team Confluence Support Page: SonarQube https://sonarqube.allstate.com/sessions/new?return_to=%2F
5	Artifactory	**Artifactory** is a product by JFrog that serves as a binary repository manager. ... The binary repository is a natural extension to the source code repository, in that it will store the outcome of your build process, often denoted as artifacts.	Team Confluence Support Page: Artifactory https://artifactory.allstate.com/artifactory/webapp/#/home
6	Pivotal Cloud Foundry (PCF)	**Pivotal Cloud Foundry** is an open source, multi-cloud application platform as a service governed by the Cloud Foundry Foundation, a 501 organization. The software was originally developed by VMware, transferred to Pivotal Software but brought back into VMware at the end of 2019 with VMware's take over of Pivotal.	Team Confluence Support Page: Pivotal Cloud Foundry(PCF)
7	Conveyor	Conveyor is an Allstate-specific application that orchestrates compliance, security, and change management within the deployment process for multiple platforms. Simple setup process Automates ServiceNow change records* Validates and updates change records	https://docs.compozed.allstate.com/pages/deploying_apps/deploying_with_conveyor.html BaroMeterID Requirement: https://allstate.barometerit.com/b/saml/efd86dae2aa04abb8a1b38c27eba84ac

		Automatic security scans through Agent Smith Automatic compliance checks through Compliance Buddy Manages change freeze Facilitates automated CMDB updates* Automates creating ServiceNow configurable items* Automatic rollback on deployment failures* Concurrently pushes your application to "platform"* Convenient Jenkins plugin Blue green deployments* To support migration of PCF environments to AWS, changes are required to ensure BarometerIT number is known during application deployment. BarometerIT number will be used to lookup application risk classification required to provision to the proper AWS environment i.e. high, moderate, etc. **Effective April 4**, any deployment that cannot be traced back to a valid BarometerIT number will fail due to missing BarometerIT number input. Conveyor will first attempt to lookup the BarometerIT number associated to PCF organization. Reference DOCS to check and/or update this relationship as this is the preferred method for accurate configuration management.	
8	JWT	JSON Web Token (JWT) is an open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the HMAC algorithm) or a public/private key pair using RSA or ECDSA. Although JWTs can be encrypted to also provide secrecy between parties, we will focus on signed tokens. Signed tokens can verify the integrity of the claims contained within it, while encrypted tokens hide those claims from other parties. When tokens are signed using public/private key pairs, the signature also certifies that only the party holding the private key is the one that signed it.	https://jwt.io/
9	Compliance Buddy	Reduce audit and security touch points by providing a centralized location with early risk identification and compliance automation. Compliance Buddy allows product teams to develop and deploy with confidence knowing their product will meet audit and security standards.	https://compozed.allstate.com/index.php/compliance-buddy/ https://github.allstate.com/IS-ComplianceBuddy/ComplianceBuddy-frontend/wiki

10	Agent Smith	Agent Smith provides a way to do application security testing by integrating with our preferred third party scanning tools. Agent SMITH has automated Veracode & Blackduck scans for Teams on PCF using Conveyor.	https://compozed.allstate.com/index.php/test-767/
11	Micro-services (*Optional*)	Microservices is a technique —a variant of the service-oriented architecture structural style— that arranges an application as a collection of loosely coupled services. In a microservices architecture, services are fine-grained and the protocols are lightweight.	Domain-Based Microservices https://app.pluralsight.com/library/search?q=microservices
12	Proxy Pattern (*Optional*)	Since the Internet era, most applications that leverage an n-tier or client-server architecture prioritize isolating the client/UI and server/application tiers. With the emergence of API and microservice based architecture, client/UI and server/application tiers are increasingly independent of each other. Single Page Applications (SPAs) tend to use very different architecture style and programming methods than the server implementation. The stateless interaction between client and server has allowed the fast adoption of mobile clients via REST APIs. The API interface is the only common language between the client and server tiers. In current Agile/XP methodologies a single product team is often responsible for building building both the UI and server side components. A typical approach for building modern web applications includes building two separate client and server projects. The client side UI project is typically JavaScript and markup (HTML, CSS). The server project primarily fulfills client requests and can be built with a variety of technologies (Java, Go, Node.js, .NET, etc) and would represent API or microservices. In the CompoZed labs it is often a Spring Boot based project.	UI - Proxy Pattern
13	Node.JS node-gyp	Node.js is an open-source, cross-platform, JavaScript runtime environment that executes JavaScript code outside of a web browser Node.js® is a JavaScript runtime built on Chrome's V8 JavaScript engine.	https://nodejs.org/en/ https://github.com/nodejs/node-gyp
14	npm	npm is a package manager for the JavaScript programming language. It is the default package manager for the JavaScript runtime environment Node.js. It consists of a command line client, also called npm, and an online database of public and paid-for private packages, called the npm registry. Wikipedia	https://www.npmjs.com/ https://www.sitepoint.com/beginners-guide-node-package-manager/
15	JAVA Version 8	OO Programming Language	https://www.oracle.com/technetwork/java/javase/overview/java8-2100321.html

16	Spring Framework	The Spring Framework is an application framework and inversion of control container for the Java platform. The framework's core features can be used by any Java application, but there are extensions for building web applications on top of the Java EE platform.	https://spring.io/
17	Spring Boot	**Spring Boot** is an open source Java-based framework used to create a micro Service. It is developed by Pivotal Team and is used to build stand-alone and production ready **spring** applications.	https://spring.io/projects/spring-boot
18	Swagger	Swagger is an open-source software framework backed by a large ecosystem of tools that helps developers design, build, document, and consume RESTful web services. Wikipedia	Team Confluence Page: Swagger
19	AppFabric MarketPlace	Marketplace is a self-service portal for product teams to get started on one of Allstate's destination platforms and leverage essential services. Quickly provision your path to production and get started with the click of a button. May think of Marketplace as a Service Repository or Service Commerce Library or set of Commerce Service Libraries that may be extended, leveraged, andOr injected for application project/product use. *Marketplace uses **Symphony** to execute contributed services.* What can the Marketplace do? Create a new system record in Barometer IT A new product does not yet have a Barometer IT number (a unique ID assigned to each Allstate product). Create a System Role ID Auto-generate PCF Org Auto-generate Jenkins access and space Create Github Org and grant user's org permissions Create Artifactory repository and grant system account access	https://marketplace.mgmt.allstate.com/
		Platform as a Service (Paas)	https://marketplace.mgmt.allstate.com/onboard-paas
		For MarketPlace Contributors:	
		Symphony Services:	https://symphony-auth.mgmt.allstate.com/login **How to Onboard Symphony Services**

		Symphony is an microservice orchestration application built around Netflix's Conductor while adding security and authentication. The fundamental purpose of Symphony is allowing the execution of changes with a shorter time-to-market, improved visibility and operational agility for all of ATSV through the orchestration of workflows built on microservices.	https://symphony.mgmt.allstate.com/#/help/overview **How to Contribute to the MarketPlace** https://github.allstate.com/pages/Developer-Marketplace/Marketplace-Help/pages/contributeToMarketplace.html **How to Generate a JSON Web Key(JWK)** ***** Reference the JWT mentioned above ***** https://symphony.mgmt.allstate.com/#/help/providers#keyGen **Symphony Developer Guide:** https://developers.symphony.com/symphony-developer/docs **Allstate Example of Symphony Service:** https://github.allstate.com/pages/Developer-Marketplace/Marketplace-Help/pages/contribution/writingServices.html https://github.allstate.com/lifeguard/Sample-Service-Worker
		Netflix's Conductor Built Conductor to help orchestrate microservices based process flows	https://github.allstate.com/pages/lifeguard/conductor/
		Netflix's Zuul	https://github.com/Netflix/zuul
20	AppFabric Middleware	**AppFabric** Framework brings consistent security and other crucial, essential services to applications allowing product teams to maximize their time spent on capabilities versus setup and plumbing. Our automation tools enable teams by providing code generation for the web, service, and proxy tiers, credential management with Vault integration, pipeline configuration, and artifact registration for compliance gating. The AppFabric CLI is used to generate initial code and related artifacts for a new app. In addition, it is possible to augment existing apps. For example, an existing app can pull in and leverage the AppFabric Framework's logging and/or vault components.	https://github.allstate.com/pages/Engineering/AppFabric/
		Dependencies and Requirements	***The Java AppFabric Spring Framework is for use with applications using the Spring Framework. AppFabric integrations are incompatible with Java applications not using Spring.***

			The AppFabric CLI is written in Node.js to enable operating system portability. The following MUST be installed on any machine which will leverage the CLI. **Node.js 8.11.3 or greater** **Java 8** **For a seamless experience, the following should be available before running the CLI:** GitHub organization that is associated with the user's ntid PCF organization that is associated with the user's ntid Jenkins organization on jenkinscf.allstate.com (Jenkins org name must match PCF org name) Artifactory Repository – for build artifacts System Role id/password – only for the first use of the CLI For details on obtaining items in the above list, please refer to the FAQ - Running CLI document on this site. **Note: New applications will need to complete the onboarding process in Martketplace.** **The CompoZed website is a good resource to consult for any questions on CompoZed processes and technologies.**
		Getting Started...	https://github.allstate.com/pages/Engineering/AppFabric/docs/gettingStarted.html

The AppFabric command-line interface (CLI) is a utility designed to rapidly provide automation and security for product teams. It automatically generates code, GitHub repositories and Jenkins pipelines to automate many activities associated with Sprint 0. This will allow product teams to start writing business logic much more quickly.

Utilizing the AppFabric CLI also ensures consistency by following security standards, using standard code templates and integrating with GitHub, PCF, Jenkins, sonarqube and Artifactory.

When the AppFabric CLI is run, the default behavior will generate code for each of the selected archetypes (web, API, proxy). The following actions will be performed for each of the selected archetypes:

Generate source code
Create a new GitHub repository within the provided GitHub organization
Push the generated code to the new GitHub repository
Create two Jenkins jobs connected to the new GitHub repository:
Deployment job which performs a build on the source code and attempts to deploy to DEV, UAT, and STAGING (which will be stopped if successful)
Pull Request job that will compile and execute the tests on the PR branch

Each Jenkins job will perform the standard tasks. It will also satisfy audit and compliance needs.

Build, execute automated unit tests, and deploy the build artifact to Artifactory
Integrate with sonarqube for static code analysis
Leverage Conveyor for deployments. Conveyor will automatically integrate with Compliance Buddy and AgentSMITH, which will also initiate security code scans.

Currently, AppFabric leverages the proxy pattern architecture to set up the microservices and UI deployed services.

		Note: As described above, the AppFabric CLI is used to generate new apps. In addition, it is possible for existing Spring apps to utilize AppFabric Framework components such as the logging and vault components.